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Polyol-assisted ternary deep eutectic solvent protective lignocellulose pretreatment for high-efficiency xylan utilization and ethanol production

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ABSTRACT

The traditional lignocellulose pretreatment by deep eutectic solvent (DES) was usually conducted under higher acidic, alkaline and high temperature conditions, which leads to the severe degradation of xylan, decreasing the subsequent reducing sugar concentration by enzymatic hydrolysis and further ethanol fermentation. It is essential to develop an effective DES that selectively removes lignin while preventing excessive xylan degradation during lignocellulose pretreatment. An effective ethylene glycol-assisted ternary DES was designed to treat corn straw (CS) at 100 °C for 6 h. 65.51 % lignin removal was achieved, over 93.46 % cellulose and 50.22 % xylan were retained in pretreated CS with excellent enzymatic digestibility (glucan conversion of 77.05 % and xylan conversion of 71.72 %), total sugar conversion could reach 75.93 %, implying the unique capacity to selectively remove lignin while preserving carbohydrate components. Furthermore, the universality of the selective removal of lignin and effective retention of xylan by ternary DES has been successfully proven by other polyols. The enzymatic hydrolysate of ternary DES-pretreated CS fermented over our genetically engineered yeast strain SFA1^{OE} gave a high ethanol yield of 0.488 g/g total reducing sugar, demonstrating the effectiveness of the polyol-assisted ternary DES pretreatment in achieving high-efficiency cellulosic ethanol production.

1. Introduction

1.1. Engineering conversion of lignocellulosic biomass

The impending depletion of fossil resource and several serious environmental burdens caused by utilization of fossil fuels has gained increasing concern worldwide (Liu et al., 2021). It has been found that excessive greenhouse gas emissions mainly caused by consumption of fossil fuels, leads to many negative effects including climate change, receding of glaciers, rise in sea level, loss of biodiversity, etc. (Jones, Peters, & Gasser, 2021). As the most abundant renewable feedstock on the earth, lignocellulosic biomass consisting of polymers of cellulose, hemicellulose and lignin has received much attention for its use in the production of chemicals and fuels through biological chemical means (Cheng et al., 2023). As measured by Díez, Urueña, Piñero, Barrio, and Tamminen (2020), the composition of corn straw, is 33–38 % cellulose, 26–32 % hemicellulose and 17–19 % lignin. Both cellulose and hemicelluloses could be able to convert into fermentable sugars. In this regard, high yields of cellulose and hemicellulose should be retained while lignin which are hard to be subsequently utilized should be removed. However, previous work has shown that in the process of enzymatic hydrolysis of lignocellulose, the non-productive adsorption of enzyme onto lignin caused by the hydrogen bonding, hydrophobic, and electrostatic interactions are thought to be the main reasons for the low enzymatic efficiency of lignocellulose (Yarbrough, Mittal, Mansfield, et al., 2015). The enzyme protein folded structure is lost during the adsorption, thus decreasing the enzymatic activity. In addition, Steric hindrance of lignin macromolecule and lignin-based derivatives produced during the pretreatment process also shows negative effects on subsequent enzymatic hydrolysis (Yuan, Jiang, Chen, et al., 2021). In the plant cell wall, lignin also linked to hemicellulose via various alkyl/aryl-ether bonds to form the lignin-carbohydrate complexes that blocks

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the access of enzymes to cellulose during enzymatic hydrolysis (Huang et al., 2022). Given this, pretreatment as the key and initial step in a lignocellulosic biorefinery is often performed to fractionate the lignocellulose components, crack the lignin-polysaccharide bonds to remove lignin from biomass and enhance the accessibility of fermentable sugars (Zhang, Xia, & Ma, 2016).Various pretreatment methods including ball milling, laccase, steam explosion, supercritical fluids, dilute acid, alkaline, and organic solvents have been applied to overcome the recalcitrance of lignocellulosic biomass feedstock (Liu et al., 2021). However, these methods have not fully addressed the issues of biomass conversion efficiency and economic feasibility (Zhao et al., 2021).

1.2. Composition and preparation deep eutectic solvents

Recently, ionic liquids (ILs) have been extensively exploited for biomass pretreatment due to their excellent ability to destroy the crystalline structure of cellulose or remove lignin. Deep eutectic solvents (DESs) are acknowledged as advanced ILs because they share similar characteristics with ILs. While common imidazolium- and pyridine-based ionic liquids are costly and cumbersome to prepare, DES can be prepared from natural components by simple heating and mixing, which has obvious cost and environmental advantages (Azzouz & Hayyan, 2023; Durand, Lecomte, & Villeneuve, 2013). DESs are commonly prepared by mixing hydrogen-bond acceptors (HBAs) and at least one hydrogen-bond donor (HBD), during which self-association of HBAs and HBDs occurs by strong hydrogen-bonding interactions. This modularity allows the chemical properties and functionality of DESs to be customizable by different combinations of HBAs and HBDs (Randelli, 2009). The most commonly used preparation method involves heating and stirring the constituents of the DES together under an inert atmosphere until a homogeneous liquid is formed (Hansen et al., 2021).

Benefiting from their high selectivity for lignin extraction, DESs can selectively remove lignin, retain cellulose and hemicellulose during the treatment process and provide opportunities for and efficient saccharification of cellulose and hemicellulose. In addition, DESs has the advantages of excellent recyclability. In the research of Ci et al. (2020) and Cheng et al. (2023), DESs can be used multiple times without significantly reducing its efficiency, and its structure and properties remain virtually unchanged throughout the cycle.

1.3. Mechanism of pretreatment of lignocellulosic feedstock using DES

Free protons in DESs were one of the critical factors in the process of biomass delignification (Xu, Kong, Peng, Wang, & Li, 2021). The interaction between DESs and biomass may be explained as an acid–base catalysis mechanism. Under the catalysis of the strong H-bond between HBD and HBA, the solvent is strong enough to compete with the inner linkages in biomass and leads to the cleavage of the labile ether linkages between phenylpropane units of lignin and the hydrogen bonds and covalent bonds between lignin and hemicellulose (Xia et al., 2018). For this reason, DESs are able to exceptionally destroy the crystal structure of cellulose or dissolving lignin. With the chloride anion as the active component, choline chloride (ChCl) become a commonly HBA, preforms a great cleavage rate of β-O-4 and thereby increases the delignification rate of biomass (Smink, Juan, Schuur, & Kersten, 2019). It has been report that ChCl:lactic acid, performed well in biomass pretreatment (Kwon et al., 2020).

1.4. Novel glycol-assisted ternary DES

Previous studies have confirmed that the enhanced delignification and subsequent enzymatic hydrolysis efficiency were found to be related to the acid amount, acid strength and the nature of hydrogen bond acceptors (Zhang et al., 2016). Liang et al. (2021) found that hydrogen bond acidity (Kamlet-Taft α parameter) had strong positive relationships with pretreatment efficacy of the studied DESs through synthesizing a series of LA-based DESs with ten different structured HBAs and applied them to pretreat corn stover (CS). However, structural instability of hemicellulose under acidic conditions leads to a significant loss of its value in the pretreatment process. Meanwhile, xylan as the dominant component in hemicellulose is inclined to degrade into low molecular weight compounds and reducing monosaccharide (e.g., oligosaccharides, xylose) in acidic DESs, making xylan recovery via precipitation almost impossible. In the strongly acidic DES system, the reducing monosaccharides hydrolyzed from xylan were able to further degrade by dehydration to form furfural, which caused problems in subsequent product separation and DESs recycling (Chen, Jacoby, & Wan, 2019; Guo et al., 2019). Through incorporating ethyleneglycol (EG) in ChCl: oxalic acid (OA) to improve active stabilization of intermediates formed during lignocellulosic biomass pretreatment, Liu et al. (2021) designed a ternary acidic DES (ChCl:OA:EG) for efficient biomass fractionation and stabilization of solubilized lignin. Therefore, DESs in ternary systems may have an advantage over similar binary systems due to the more selectable types and adjustable ratios of HBDs and HBAs. Developed by existing DES for ternary systems, several double-HBDs DESs which contains choline chloride: lactic acid: ethyleneglycol (ChCl:LA:EG), choline chloride: lactic acid: polyethylene glycol (ChCl:LA:PEG) and Choline chloride: lactic acid: glycerol (ChCl:LA:GLY) were synthesized for pretreatment. We demonstrated the optimal ratio and optimized pretreatment conditions of the ternary system ChCl:LA:EG. In addition, we tried to analyzed the mechanism of pretreatment of lignocellulose with ChCl:LA:EG with means of SEM, FTIR, TG and DSC. In this paper, a novel lignocellulose pretreatment strategy with high efficiency and low cost, which provides a novel, simple and feasible saccharification process for lignocellulose biorefinery, were proposed.

We focused on the efficiency of DES pretreatment toward improving enzymatic digestibility of lignocellulosic biomass, which is a key line in utilizing lignocellulosic raw materials for efficient ethanol fermentation. To this end, we analyzed the contribution of the three DNSs to the subsequent enzymatic hydrolysis by reducing sugar yield.

2. Materials and methods

2.1. Materials

The dried corn straw (CS) was obtained from Donghai County, Jiangsu Province, China. The CS were mechanically ground into particle size of 60 mesh and pre-extracted with toluene/ethanol (2:1, *v*/v) at 110 °C for 6 h, to remove wax. Choline chloride (ChCl), lactic acid (LA), and ethylene glycol (EG) were purchased from Macklin Chemical Co., Ltd. (Shanghai, China). Glycerol (GLY) and polyethylene glycol (PEG) were purchased from Aladdin Biochemical Co., Ltd. (Shanghai, China). Filter paper enzyme activity of 140 FPU/mL of cellulase (Novozymes Cellic CTec 3) was purchased from Sigma-Aldrich Co., Ltd. (Shanghai, China).

2.2. DESs preparation and CS pretreatment

Three ternary DESs (ChCl:LA:EG, ChCl:LA:GLY, and ChCl:LA:PEG) were prepared by mixing ChCl, LA, and polyol (EG, GLY, and PEG) in the designed molar ratios of 1:1:1, 1:1:3, 1:2:2, 1:3:1, 1:4:4, respectively, following by heating and stirring at 80 °C until a transparent and homogenized liquid was formed. Two traditional binary DESs (ChCl:EG and ChCl:LA) were prepared in the same way as for the preparation of ternary DESs, and set the same HBA molar content as ternary DESs, with the molar ratios of HBA to HBD were 1:2, 1:4, and 1:8, respectively. The CS pretreatment was carried out at 100 °C for 6 h with a solid-liquid ratio of 1:10 (5 g CS in 50 g DESs), and three sets of parallel experiments were set up for each type and proportion of DESs. After pretreatment, the slurry was separated by vacuum filtration and washed with anhydrous ethanol to remove DESs until the filtrate is neutral. Subsequently, the washed CS was dried in an oven at 80 °C and

recovered for further analyses. Solid recovery, xylan removal efficiency, and lignin removal efficiency were calculated based on Eqs. (1)–(3), respectively:

$$\text{Solid recovery}\ (\%) = \frac{\text{Pretreated CS weight}\ (g)}{\text{Raw CS weight}\ (g)} \times 100 \quad (1)$$

$$\text{Xylan removal}\ (\%) = \left(1 - \frac{\text{Xylan in pretreated CS}\ (g)}{\text{Xylan in raw CS}\ (g)}\right) \times 100 \quad (2)$$

$$\text{Lignin removal}\ (\%) = \left(1 - \frac{\text{Lignin in pretreated CS}\ (g)}{\text{Lignin in raw CS}\ (g)}\right) \times 100 \quad (3)$$

2.3. DESs recovery and reuse

Recovery of DESs required the removal of as many of the dissolved fractions as possible. After pretreatment and separation, the filtrate fraction was concentrated by rotary evaporation and then poured into deionized water to remove the dissolved lignin. Subsequently, the regenerated DESs were obtained by filtration and concentration. The recovered DESs were further dried under vacuum at 70 °C for 24 h. Next, the recovered DESs were reused to pretreat CS at 100 °C for 6 h with a solid-liquid ratio of 1:10 (5 g of CS in 50 g of DESs). After the pretreatment, the slurry was separated by vacuum filtration and washed with anhydrous ethanol. Subsequently, the washed CS was dried in an oven at 80 °C and recovered for further analysis. The lignin and xylan removal efficiency during each recycling process were examined to evaluate the DESs recycling performance.

2.4. Enzymatic hydrolysis and fermentation

$$\textit{Glucan conversion} = \frac{\textit{glucose in the hydrolysate}\ (g) \times 0.9}{\textit{cellulose in the pretreated CS}\ (g)} \times 100\%$$

$$\textit{Xylan conversion} = \frac{\textit{xylose in the hydrolysate}\ (g) \times 0.88}{\textit{xylan in the pretreated CS}\ (g)} \times 100\%$$

An engineered yeast strain SFA1^{OE} fermenting glucose and xylose was applied for ethanol fermentation evaluation. Yeast cells from single colony were firstly inoculated in 100 mL of liquid YPD (10 g/L yeast extract, 20 g/L peptone and 20 g/L dextrose) in a 250-mL conical bottle and incubated at 30 °C with 200 rpm shaking for 36 h. When yeast cells reached an optical density at $OD_{600} = 1$, they were used to inoculated 100 mL of the prepared CS hydrolysates. All fermentations were conducted under the anaerobic condition at 30 °C with 150 rpm shaking, and each fermentations sample were performed in triplicate. Samples were collected at different time points for the HPLC analysis.

2.5. Analytical methods

The composition of CS and all DESs pretreated CS samples were determined by the National Renewable Energy Laboratory protocols (Sluiter et al., 2008). The FT-IR spectra, the morphology of CS and that of the pretreated CS were analyzed by Tensor II spectrometer (Bruker, Germany) and field emission scanning electron microscope (FE-SEM, SU8010, Japan), respectively. The crystallinity index (CrI) of CS before and after pretreatment was calculated using the peak intensity method. Brunauer-Emmett-Teller (BET) analysis was conducted to estimate the accessibility of pretreated CS according to previous method (Ma et al., 2021), which were analyzed by a Kubo-X1000 surface area analyzer (Builder Electronic Technology co., Ltd., Beijing, China). Thermogravimetric analyzer STA200 (Hitachi, Japan) was applied to investigate the thermal stability of CS, pretreated CS, and prepared DESs in nitrogen atmosphere. The melting point of prepared DESs was tested on a differential scanning calorimetry 200F3 (TA Instrument, America) with a heating rate of 10 °C·min^{-1}. The Kamlet-Taft parameters of prepared DESs were measured on a UV–vis spectrophotometer (UH4150, Hitachi) and calculated by the equations of previous publication (Wang et al., 2023).

2.6. Statistics analysis

Quantitative data was described in the form of means ± standard deviation, such as the error bars in the figures. At least three samples were conducted to test the average components contents, lignin and xylan removal, and sugar yields.

3. Results and discussion

3.1. Effectiveness of ethylene glycol-assisted ternary DES in the lignocellulose pretreatment

To explore the effectiveness of ethylene glycol-assisted ternary DES for delignification and hemicellulose preservation during lignocellulose pretreatment, the changes in the chemical composition and the enzymatic digestibility of pretreated-CS were estimated at the same conditions (100 °C, 6 h, 10 % solid loading). Initially, ternary DES systems with low melting point (< −95 °C) and high thermal stability (> 100 °C), ChCl:LA:EG DESs were successfully prepared (Fig. S1). In order to investigate the differences in the effects of the components of ternary DES on the chemical composition and enzymatic digestibility of lignocellulose, ternary DES with different molar ratios were used for lignocellulosic pretreatment (Fig. 1). With the molar ratio of ChCl:LA:EG changed from 1:1:3 to 1:3:1, xylan removal significantly increased from 37.7 % to 59.3 %. However, lignin removal was first significantly enhanced from 51.8 % to 65.5 % at 1:2:2 and then only minimally elevated at 1:3:1 (66.5 %) (Fig. 1a_1). In addition, the content of choline chloride showed similar results for the removal of lignin and xylan. With the increase of ChCl ratio in ternary DES, the xylan removal increased gradually from 42.9 % to 58.9 %, while lignin removal did not increase further with ChCl (Fig. 1a_2). These indicated that increasing the percentage of lactic acid and choline chloride had a more prominent effect on the dissolution of xylan than lignin. Therefore, considering the simultaneous achievement of efficient lignin removal and xylan preservation, 1:2:2 was selected as the molar ratio of the ternary DES used for protective lignocellulose pretreatment. Thanks to the high retention of xylan and high removal of lignin after ChCl:LA:EG (1:2:2) pretreatment, the excellent enzymatic digestibility (glucan conversion of 77.05 % and xylan conversion of 71.72 %) and the highest total sugar conversion (75.93 %) could be achieved (Fig. 1b_1, b_2). Considerable highly efficient carbohydrates recovery (93.46 % cellulose, 50.22 % xylan) and 65.52 % lignin removal were achieved by ethylene glycol-assisted ternary DES pretreatment, implying the unique capacity to selectively remove lignin while preserving carbohydrate components. Compared with some traditional pretreatment methods (Cao et al., 2014; Guo et al., 2019; Liang et al., 2021; Ma et al., 2021; Mahmoodi, Karimi, & Taherzadeh, 2018; Wang et al., 2022; Wang, Long, Zhao, & Li, 2024), this ternary DESs as a highly efficient recyclable have greater selectivity for lignin dissolution, while better retaining glucan and xylan, thus ensuring that more oligomerized products and monosaccharides were produced in the subsequent enzymatic hydrolysis process, thus inducing a higher glucan and xylan conversion rates. High glucan and xylan conversion rates means more production of monosaccharides (e.g. glucose, xylose) and oligomers, compounds that are more easily utilized by engineered microorganisms as a carbon source and converted into high-value secondary metabolites products such as bioethanol. This glycol-assisted ternary DES pretreatment achieved high conversion of glucan and xylan and obtained high reducing sugar concentrations, laying the groundwork for achieving the complete utilization of the whole fraction of carbohydrate polymers in lignocellulose. In addition, an industrial grade reducing sugar concentration provides the basis for the industrial preparation of bioethanol (Table S1).

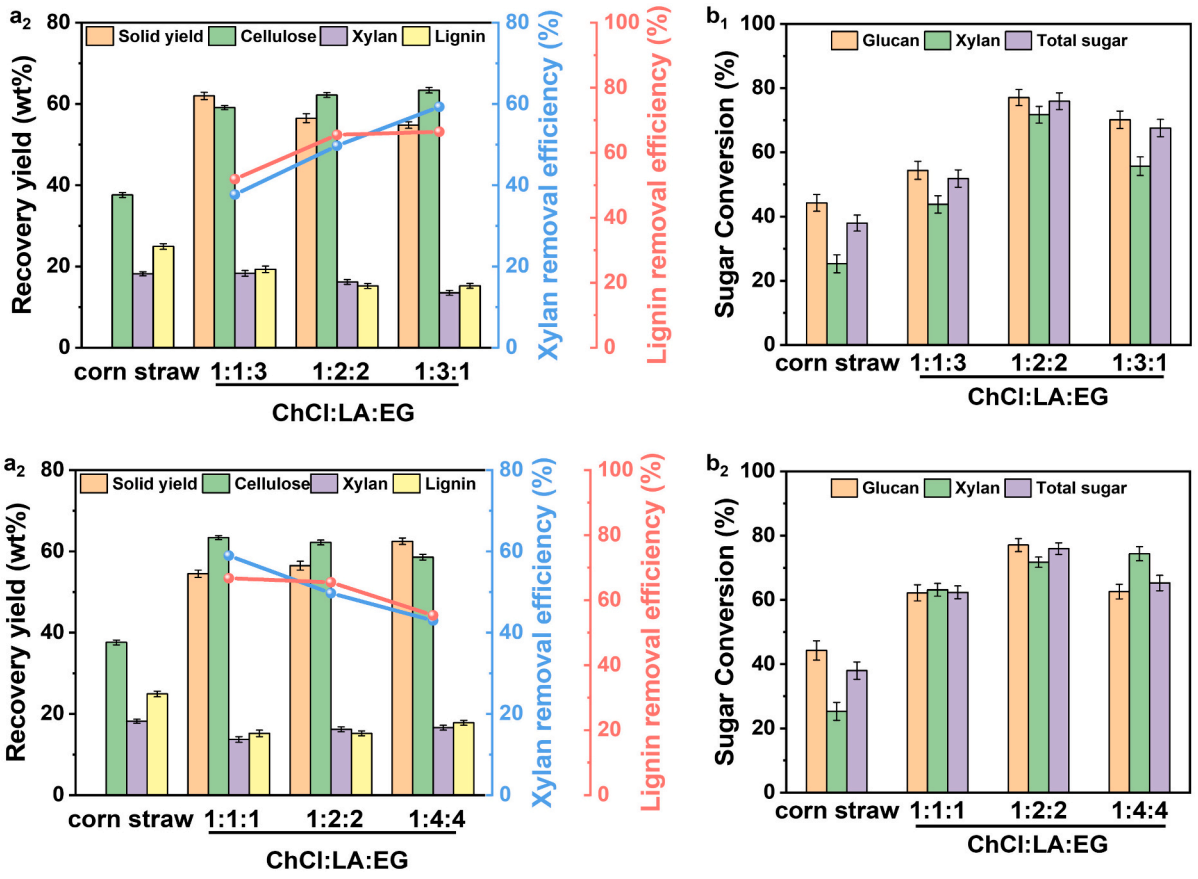


Fig. 1. Pretreatment efficiency by ethylene glycol-assisted ternary DESs with different molar ratios (a_1, a_2) and the effect of molar ratios of ternary DESs on the enzymatic digestibility of pretreated CS (b_1, b_2). a_1, a_2: recovery yield of pretreated CS and lignocellulosic components, lignin and xylan removal efficiency obtained after ethylene glycol-assisted ternary DES pretreatment under same conditions (100 °C, 6 h, 10 % solid loading). b_1, b_2: sugar conversions of pretreated CS at 10 % solid loading for enzymatic hydrolysis.

In contrast, compared with the traditional binary ChCl:LA system, as the ratio of lactic acid to choline chloride increased from 2:1 to 8:1, xylan removal improved from the 76.4 % to 84.4 % and no significant change in the lignin removal. In the same conditions tested, the efficiency of ethylene glycol-assisted ternary DES synthesized with a molar ratio of 1:2:2 gave lignin removal (65.51 %) comparable to that of ChCl:LA (~ 65 %), whereas xylan removal (49.78 %) was significantly inhibited in comparison (Fig. 2a_2). The differences in the dissolution of lignocellulosic components in the pretreatment could be associated with the Kamlet-Taft empirical parameters of the solvents. This empirical model characterized solvents by three parameters: polarity (π*), H-bond donating ability or acidity (α) and accepting ability or basicity (β) (Liu et al., 2019). The DES with relative higher values of α generally showed better lignin extraction and dissolution efficiency during pretreatment (Yu, Xue, Shi, Zhou, & Mu, 2022). As shown in Table S2, ChCl:LA (1:4) had a highest α value (1.29) could easily donate H-bonds to release H^+, which could efficient cleavage of lignin-carbohydrate complex bonds, ether bonds of lignin, and glucosidic bonds of xylan, achieving high lignin and xylan removal efficiency (Liu et al., 2017). With the addition of EG to ChCl:LA decreased the α value of ChCl:LA:EG, which weakened the ability to disrupt glycosidic linkages of xylan, thus preserving the xylan from excessive degradation by ChCl:LA:EG. However, the weakened H-bond acidity still provided a good removal efficiency for lignin, indicating that the ether bonds of lignin was more readily ruptured than the glycosidic bonds of xylan in DES pretreatment. Base on the above results, polyol could effectively optimize the H-bond acidity of DES to achieve selective lignin removal while weakening excessive xylan degradation. Due to the high lignin removal and the conservation of xylan, compared to all pretreated CS obtained by traditional binary DES (ChCl:EG, ChCl:LA) pretreatment, pretreated CS obtained by ternary DES (ChCl:LA:EG) pretreatment displayed superior sugar conversion (Fig. 2b_1, b_2). The highest total sugar conversion of 75.93 % for ChCl:LA:EG (1:2:2)-pretreated CS was achieved, with glucan and xylan conversions of 77.05 % and 71.72 %, respectively. In comparison, ChCl:LA (1:4) and ChCl:EG (1:4) pretreated CS had total sugar conversions of 49.49 % and 27.95 %, with 44.12 % and 28.55 % glucan conversions, and 48.05 % and 26.46 % xylan conversions, respectively. Furthermore, the ChCl:LA exhibited stronger pretreatment efficiency and resulted in higher sugar conversion than the ChCl:EG. Therefore, ethylene glycol-assisted ternary DES with flexible and tunable H-bond acidity, exhibiting an outstanding ability for xylan retention and lignin removal during lignocellulose pretreatment.

In summary, the better pretreatment efficiency and total sugar conversion were achieved through ethylene glycol-assisted ternary DES pretreatment, which integrated the advantages of traditional binary DESs (high lignin removal by ChCl:LA and well carbohydrates preserve by ChCl:EG). As a result, 65.51 % lignin removal was achieved after pretreatment, and over 93.46 % cellulose and 50.22 % xylan were retained in pretreated CS with excellent enzymatic digestibility (glucan conversion of 77.05 % and xylan conversion of 71.72 %), total sugar conversion could reach 75.93 %. It was worth noting that ethylene glycol-assisted ternary DES pretreatment demonstrated excellent performance in carbohydrates reservation and lignin removal. Furthermore, the universality of polyol-assisted ternary DESs for carbohydrate protective pretreatment were assessed from both chemical composition and enzymatic digestibility of pretreated CS (Fig. 3a, b). With the same pretreatment conditions (100 °C, 6 h, 10 % solid loading) used in ChCl:LA:EG pretreatment, ChCl:LA:GLY and ChCl:LA:PEG pretreated CS

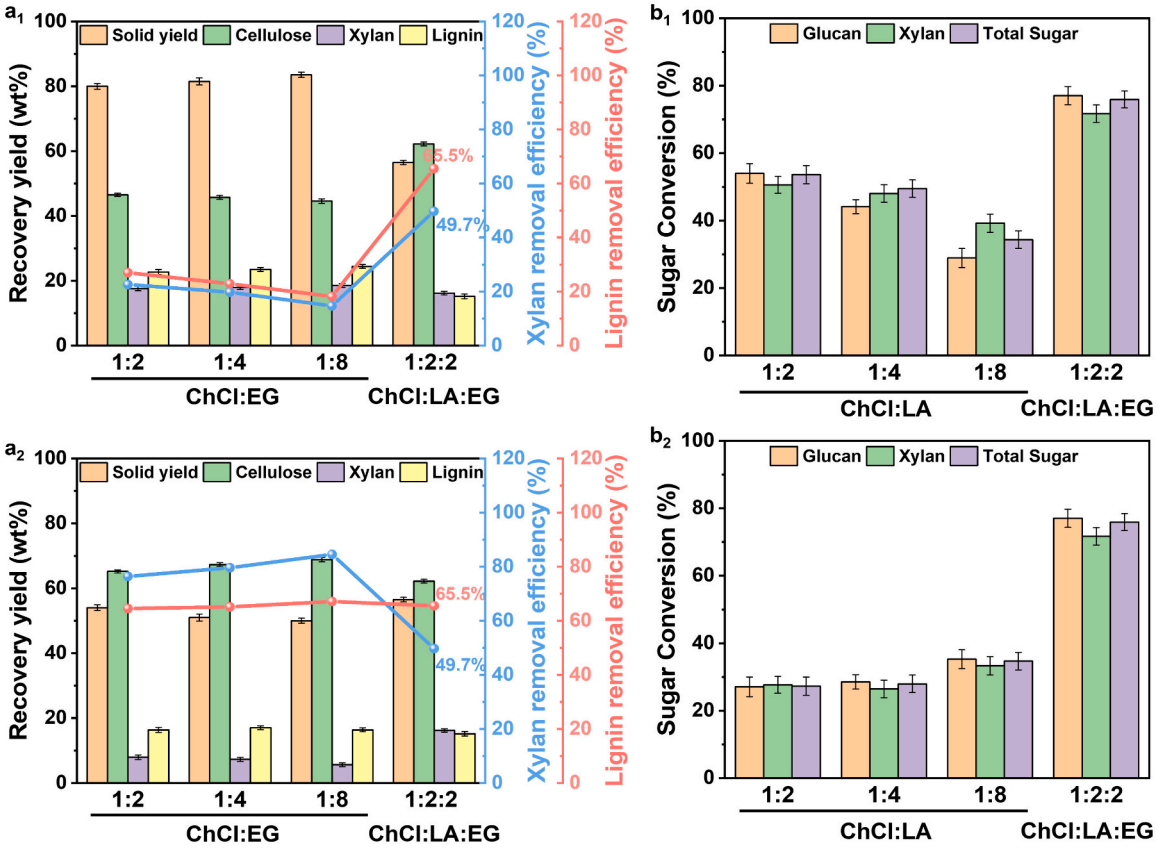


Fig. 2. Comparison of the effects of traditional binary DESs (ChCl:LA, ChCl:EG) and ethylene glycol-assisted ternary DES on pretreatment efficiency (a_1, a_2) and enzymatic digestibility of pretreated CS (b_1, b_2). a_1, a_2: recovery yield of pretreated CS and lignocellulosic components, lignin and xylan removal efficiency obtained after binary and ternary DES pretreatment under same conditions (100 °C, 6 h, 10 % solid loading). b_1, b_2: sugar conversions of pretreated CS at 10 % solid loading for enzymatic hydrolysis.

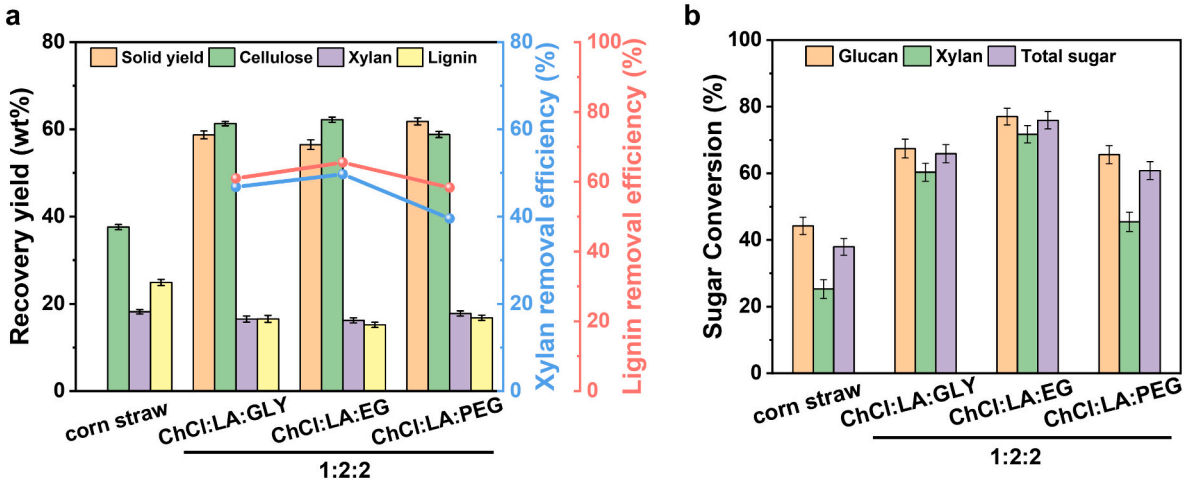


Fig. 3. Pretreatment efficiency by different polyol-assisted ternary DESs (a) and the enzymatic digestibility of CS with corresponding pretreatment.

achieved a satisfactory lignin removal of over 60 % while maintaining a comparable xylan content to ChCl:LA:EG pretreated CS. ChCl:LA:GLY and ChCl:LA:PEG pretreated samples reserved 47.3 % and 60.2 % of xylan, respectively, which was significantly lower than the xylan content of the sample from the ChCl:LA (1:4) pretreatment (20.4 %) (Fig. 3a). Meanwhile, relatively high total sugar conversions of 65.89 % and 60.84 % were obtained by ChCl:LA:GLY, ChCl:LA:PEG pretreatment, respectively (Fig. 3b). It was reasonable to believe that this ternary DES pretreatment could be tailored to perverse carbohydrates and selective lignin removal with different polyol-assisted acidic DES.

Simple recyclability was a vital feature of DESs, which was an important basis for the development of sustainable biorefinery. In order to verify the economic applicability of the pretreatment in this study, the reusability of the ChCl:LA:EG (1:2:2) was investigated at same pretreatment conditions as shown in Fig. S2. The recovery yield (%, *w*/w) of recycled DESs were at least 90 % each cycle. In the fourth cycle, lignin removal efficiency slightly decreased from 65.5 % to 56.2 %, which could be attributed to the accumulation of dissolved lignin and

hemicellulose fragments in recovered DESs (Wang et al., 2022). Furthermore, the ethylene glycol-assisted ternary DES pretreatment could still protect the hemicellulose in a satisfactory yield in the recycling experiment. The 53.1 % of xylan was reserved in the pretreated sample of the last cycle, which was significantly lower than the xylan content of the sample from the ChCl:LA (1:4) pretreatment (20.4 %). Overall, this study clearly demonstrated that the ethylene glycol-assisted ternary DES presented reusability in lignin removal and carbohydrates preservation during pretreatment.

3.2. Enzymatic digestibility of ethylene glycol-assisted ternary DESs pretreated CS at high solid loadings

To prepare industrial ethanol, it is necessary to perform enzymatic hydrolysis on the pretreated residue. Through component analysis, we screened the optimal treatment for each DES system. Enzymatic hydrolysis was carried out for 72 h at an enzyme loading rate of 20 FPU/g and a solid-liquid ratio of 10 %. From the enzyme kinetic curve (Fig. 4a), it can be seen that the enzyme digestion rate of the ternary system ChCl/LA/EG reached 75.9 %, which is much higher than the other groups. To obtain large-scale industrial ethanol, high solid biomass load is considered a prerequisite for industrial application. To obtain a better enzymatic hydrolysis effect, the concentration of reducing sugars at 10 %, 15 %, and 20 % different solid-liquid ratio was explored (Fig. 4b, c). The lower figure studies the release concentrations of glucose and xylose at 72 h after enzymatic digestion of CS. Samples were pretreated with DES and enzymatically digested at 10 %, 15 %, and 20 % of different solid loads. CS reducing sugar concentration compared with no DES pretreatment group, 10 % solid load control group have significantly increased, with the increase of solid-liquid ratio, reducing sugar concentration significantly, at 15 % of the total sugar concentration over 100 g/L, but when the solid load increased to 20 %, the reducing sugar concentration reduced to 44.74 g/L, this is when the solid load increase is too large, will lead to the decrease of water content in the enzymatic system, thus affect the rheology of pulp. Increased viscosity and incomplete mixing of cellulase with the substrate lead to uncomplete enzymatic digestion, thus reducing sugar production (Zhao et al., 2024).

In this work, 65.51 % lignin removal was achieved, over 93.46 % cellulose and 50.22 % xylan were retained in pretreated CS by ternary DES with excellent enzymatic digestibility (glucan conversion of 77.05 % and xylan conversion of 71.72 %), total sugar conversion could reach 75.93 %, implying the unique capacity to selectively remove lignin while preserving carbohydrate components. The high concentration of enzymatic hydrolysate from DES-treated residues has potential for large-scale production of cellulosic ethanol (Fig. 5).

3.3. Evaluating the structural properties of pretreated substrates

The above results indicated that outstanding capability of polyol assisted DESs pretreatment in lignocellulose fractionation, preservation of polysaccharides, and enhancement of enzymatic digestibility. The major factors responsible for the increased enzymatic digestibility of ternary DESs-treated CS should be analyzed in terms of the changes in the structural and the properties after pretreatment.

The surface morphological properties of the original and pretreated CS were unveiled by SEM analysis (Fig. S3). It was observed that the corn stover was sticks with regular neat surface stripes and relatively smooth microscopic surfaces, which thoroughly hindered the

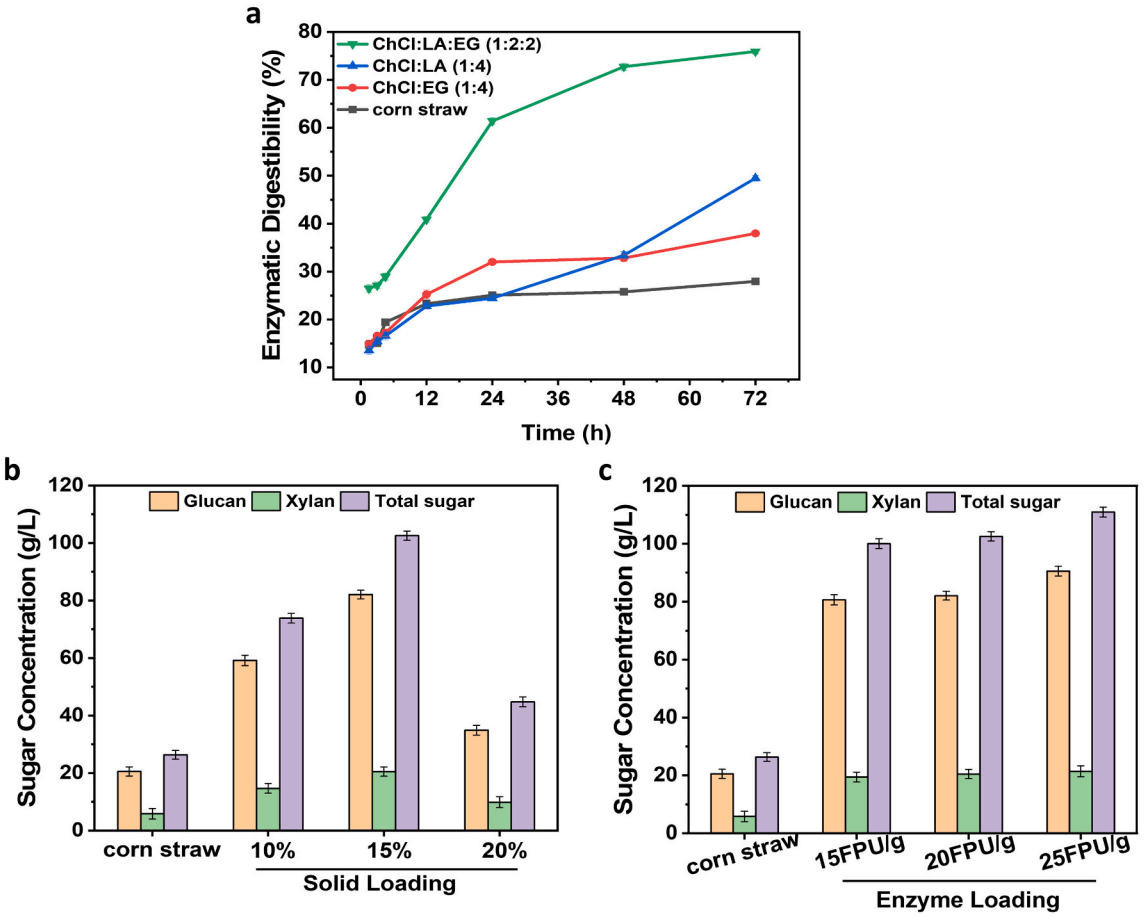


Fig. 4. Enzymatic digestibility curves of pretreated CS by different DESs systems (a), enzymatic hydrolysis of ChCl:LA:EG (1:2:2) pretreated-CS at solid loadings of 10 %, 15 % and 20 % (b) and with varying enzyme dosages (c). The enzyme dosage in (b) was 20 mg g^{-1} glucan. The solid loading in (c) was 15 %.



Fig. 5. Schematic diagram for the preparation of cellulosic ethanol from corn straw by enzymatic hydrolysis and fermentation after ternary DESs pretreatment.

accessibility of cellulase to cellulose and thus was not conducive to further saccharification, whereas the microscopic morphology of the pretreated samples was changed to varying degrees. Specifically, no visible differences could be regarded as significant between the pretreated samples by ChCl:EG (1:4) and original CS, suggesting that ChCl:EG (1:4) pretreatment was less destructive to the dense structure of lignocellulose. In contrast, ChCl:LA (1:4)-mediated and ChCl:LA:EG (1:2:2)-mediated pretreatment had resulted in a considerably disrupted

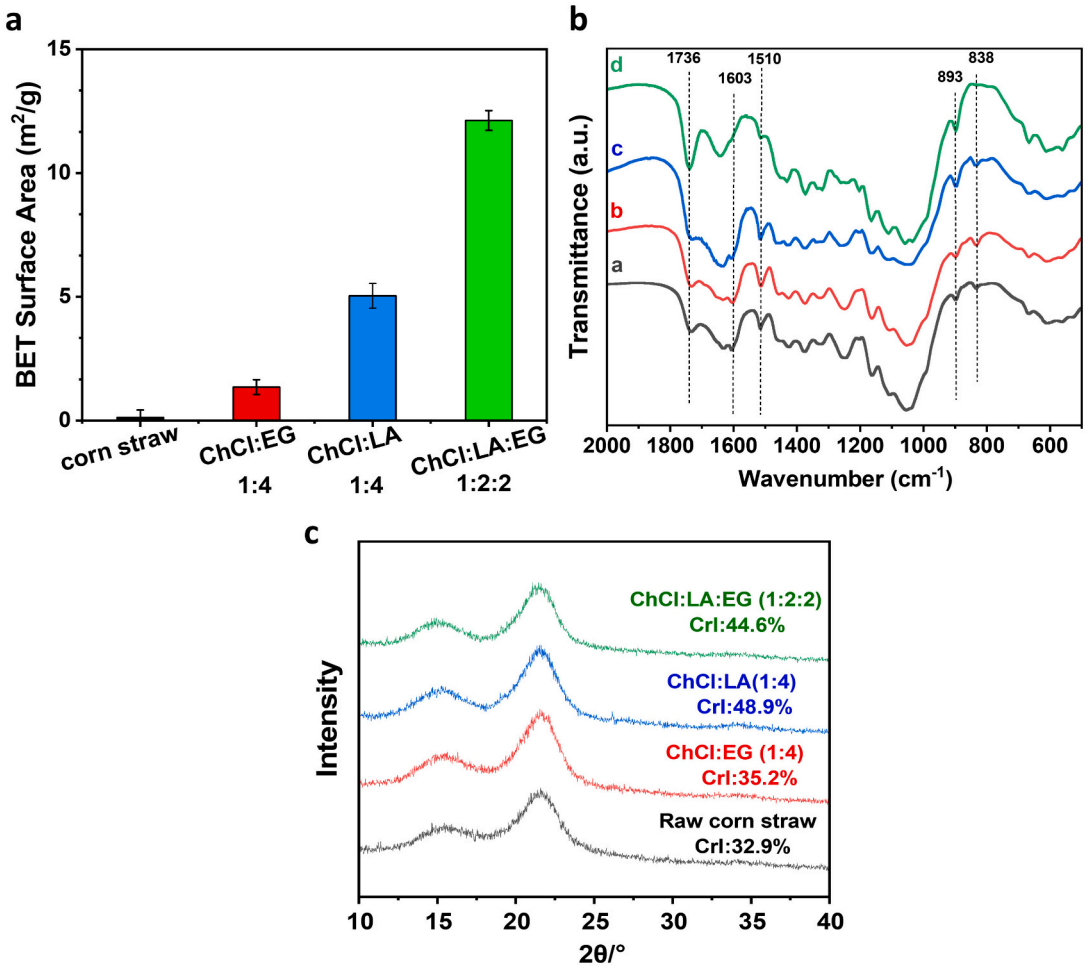


Fig. 6. BET surface area (a), FTIR spectra (b), and XRD pattern of untreated CS and pretreated CS by different DESs. a, b, c, d in the FTIR represented the raw CS and the residue after pretreatment with ChCl:EG (1:4), ChCl:LA (1:4), ChCl:LA:EG (1:2:2), respectively.

structure, exhibiting obvious cracks, porosity, fragmentation, and low fiber-bundle integrity. It was worth noting that the ChCl:LA:EG (1:2:2)-pretreated sample have a looser surface than the ChCl:LA (1:4)-pretreated sample, which might be attributed to the extreme removal of lignin and hemicellulose after the ChCl:LA (1:4)-pretreatment resulting in the collapse of the cellulose spatial structure, which would reduce the cellulase reaction sites. These morphological changes were mainly due to the significant removal of lignocellulose components (mainly lignin that greatly limited enzyme contact with the cellulose surface by nonproductive adsorption), generating large amounts of reaction sites on the fiber surface, thereby increasing the accessibility of cellulase and enhancing the further enzymatic saccharification (Xu et al., 2024). In order to demonstrate more intuitively the changes in the surface topography of the lignocellulose after DESs pretreatment, the BET surface areas were also determined to explain the differences of enzymatic digestibility for differently pretreated CS (Fig. 6a). It was well-known that an increase in surface area will enhance accessibility of cellulase to cellulose fibers and improve the enzymatic digestibility (Shen et al., 2023). According to BET analysis, the ChCl:EG (1:4) pretreated sample showed slightly higher surface areas (0.76 m^2/g) as compared to that (0.12 m^2/g) of original CS. However, the surface area of the ethylene glycol-assisted ChCl:LA pretreated sample notably increased to 12.12 m^2/g, which was 2.4 times higher than the surface area of the pretreated sample by ChCl:LA (1:4) (5.04 m^2/g). The surface area results further confirmed that excessive removal of lignin and hemicellulose resulted in the collapse of the cellulose space structure, which in turn limited enzymatic saccharification by reducing the surface area of the sample. Therefore, ethylene glycol -assisted ternary DES pretreatment not only effectively protected hemicellulose, but also enabled the treated lignocellulose to have lots of enzymatic reaction sites.

The changes in the chemical composition of the solid residues after DESs pretreatment were further investigated using FT-IR analysis (Fig. 6b). Evidently, the characteristic absorption peaks detected at 1603, 1510 and 838 cm^{-1}, assigned to the stretching vibration of the C—C and C—H bonds in the lignin unit, respectively, were remarkably weakened in the ChCl:LA:EG (1:2:2)-pretreated CS, confirming the efficient lignin removal (Yin et al., 2022). The signal at 893 cm^{-1}, corresponding to the C_1—H deformation vibration for the glycosidic band in cellulose/xylan, was sharply, indicating that the richer content of carbohydrates (Sun et al., 2009). This further validated the xylan protective effect of ethylene glycol-assisted ternary DES in lignocellulose pretreatment. However, the signal for unconjugated aliphatic carbonyl structures at 1736 cm^{-1} was clearly and sharply, suggesting the occurrence of esterification between lactic acid and aliphatic hydroxyl groups in residual lignin (Zheng, Chen, Qiu, Sun, & Lin, 2023). Esterification of residual lignin in the pretreated residue could effectively increase the hydrophilicity of the substrate and inhibit the hydrophobic adsorption of lignin to the cellulase, which in turn improved the enzymatic digestibility (Xu et al., 2024).

XRD can more accurately reflect the changes in the crystallinity of cellulose in CS before and after pretreatment, and as a reference for evaluating the performance of pretreatment and enzymatic (Fan et al., 2023). The XRD patterns of all samples retained the typical diffraction peaks intensity of the crystalline cellulose I (16°and 22°), which indicated that the crystalline structure of cellulose is not disrupted during all DESs pretreatment (Fig. 6c) (Wei, Bu, Zhou, Deng, & Zhu, 2021), It was found that the lignocellulose crystallinity dramatically increased from 32.9 % to 48.9 % after ChCl:LA (1:4) pretreatment, which was mainly due to the removal of large amounts of amorphous components such as lignin and hemicellulose. In contrast, the crystallinity of the residue pretreated with ChCl:LA:EG (1:2:2) increased to 44.6 %, which was lower than the crystallinity of the residue after ChCl:LA (1:4) pretreatment (48.9 %). This was mainly attributed to the preservation of hemicellulose during the ethylene glycol assisted ternary DES pretreatment. It was coincided with an increase in xylan content and thus the enhancement of total sugar yield. However, the crystallinity value of the ChCl:EG (1:4) pretreated sample increased slightly to 35.2 %, which was attributed to the partial removal of hemicellulose and lignin, resulting in lower specific surface area (0.76 m^2/g) and lower total sugar yield (27.9 %). Based on the surface morphology and structural analyses described above, we hypothesised that the ChCl:LA:EG ternary DES pretreatment achieved high total sugar conversion by selectively removing lignin, providing lots of enzymatic reaction sites and inhibiting non-productive adsorption by cellulase, while protecting xylan from severe degradation.

3.4. Cellulosic ethanol fermentation in the enzymatic hydrolysates of ternary DESs pretreated CS

The saccharification rate of hemicellulose and cellulose is a prerequisite for achieving high-titer ethanol production (Shen et al., 2023).

To further evaluate the fermentation performance of the CS hydrolysate after proposed DES pretreatment, the selected control strain WXY70 and its engineered strain $SFA1^{OE}$ reported in our laboratory and were fermented and investigated in the highly mixed sugars of simulated corn stalk hydrolysate containing 160 g/L glucose, 60 g/L xylose and 2.5 g/L acetic acid for 48 h (Zhu et al., 2020). As shown in Fig. 7 b_1, target $SFA1^{OE}$ gave ethanol yields of 0.37 and 0.43 g/g total sugars within 36 and 48 h, respectively, compared to 0.35 and 0.41 g ethanol/ g total sugars in WXY70, which is the 84.3 % of the theoretical ethanol conversion rate at 48 h. This suggested that $SFA1^{OE}$ consistently performed better than WXY70 during the fermentation conditions of previous and current mixed sugars of 120 and 220 g/L (Zhu et al., 2020). Therefore, the selected strain $SFA1^{OE}$ could be used to test the efficiency of above DES pretreated CS hydrolysate as followed.

After the proposed ethylene glycol assisted-ternary DES pretreated CS enzymatic hydrolysate, a total reducing sugar concentration over 100 g/L (around 83.9 g/L glucose and 19.6 g/L xylose) were obtained. During that fermentation process, the tested $SFA1^{OE}$ only left 0.5 g/L xylose and produced the ethanol production of 48.6 g/L at 48 h, achieving the efficient ethanol yield of 0.48 g/g total sugars or 94.9 % of the theoretical maximum value (Fig. 7 b_2). This promising peak value suggested that ethylene glycol assisted-ternary DES pretreated CS hydrolysate is very effective as the green pretreatment with low inhibitors in the fermentation media.

Finaly, the material input and output results of cellulosic ethanol production from CS under the determined DES pretreatment process parameters were analyzed through material balance. Based on the mass conservation of 1000 g CS, the current maximum yield of cellulosic ethanol could be reached as 155.52 g per kilogram of CS (Fig. S4).

4. Conclusions

In this study, a ternary DES pretreatment process assisted by ethylene glycol not only achieved the selective removal of lignin with well carbohydrate components preserved, but also ensured the ideal enzymatic digestibility and fermentation. After pretreatment with ChCl:LA:EG (1:2:2) at 100 °C for 6 h, 65.51 % lignin removal was achieved, with 93.46 % cellulose and 50.22 % xylan were well preserved in pretreated CS with excellent enzymatic digestibility (glucan conversion of 77.05 % and xylan conversion of 71.72 %), total sugar conversion could reach 75.93 %. Moreover, glycerol and polyethylene glycol-assisted ternary DES exhibited the similar capacity in selectively removing lignin while preserving of carbohydrate components. The xylan yield with 53.1 % was preserved in pretreated CS after four times recycling of ChCl:LA:EG (1:2:2). For the fermentation of ethylene glycol assisted-ternary DES pretreated CS enzymatic hydrolysate (> 100 g/L), $SFA1^{OE}$ produced the high ethanol titer of 48.6 g/L, suggesting that target $SFA1^{OE}$ could maximize ethanol titer in ethylene glycol assisted-ternary DES pretreated CS hydrolysate. In summary, ethylene glycol assisted-ternary DES pretreatment achieved the selective removal of lignin with well carbohydrate components preserved, thereby increasing the enzymatic reducing sugar conversions by 26.4 % and expanding similar cellulosic

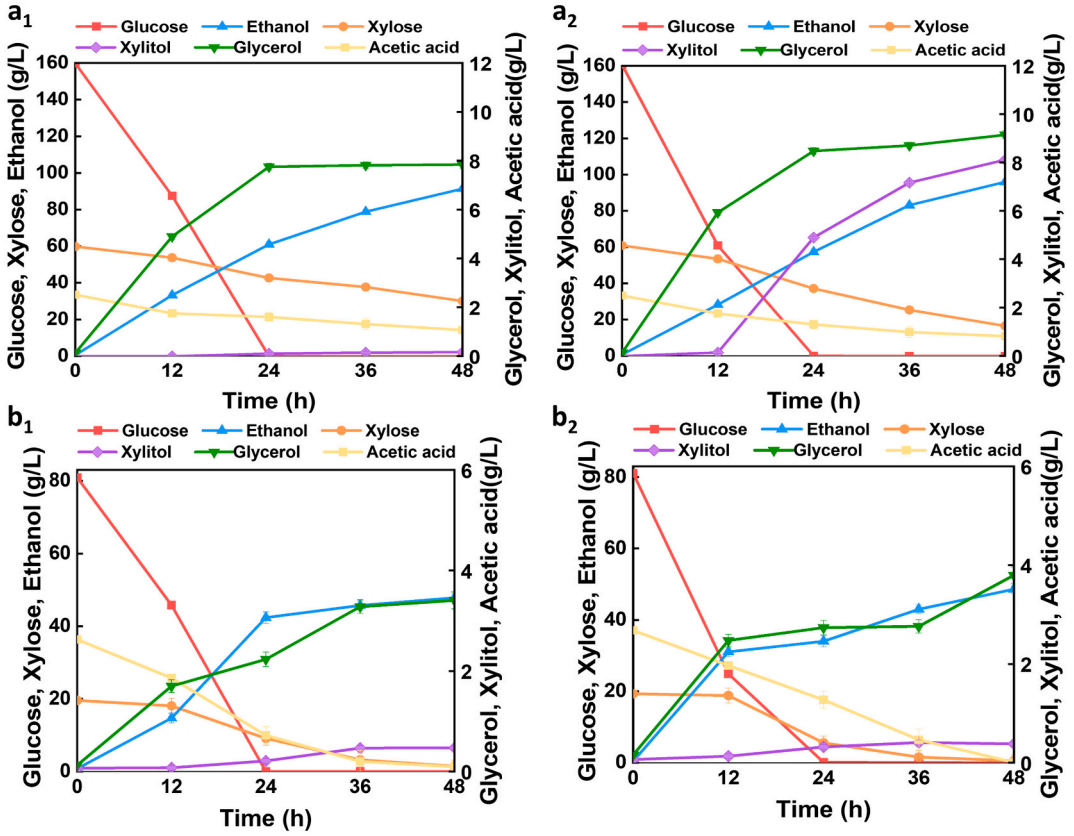


Fig. 7. Comparison of fermentation performance between control strains WXY70 and its engineered strain SFA1^{OE} under simulated high sugar corn straw hydrolysate (a_1, b_1) and the optimal condition of ChCl: LA: EG (1:2:2) pretreated CS (solid load 15 %, enzyme dosage 20 mg g^{-1}) (a_2, b_2).

ethanol yield compared to the traditional ChCl:LA.

CRediT authorship contribution statement

Shan Li: Writing – original draft, Investigation, Formal analysis. **Yang Wang:** Writing – original draft, Investigation, Formal analysis. **Qinghe Dong:** Writing – original draft. **Zitong Yuan:** Writing – original draft. **Tiancheng Mu:** Conceptualization. **Zhimin Xue:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Limin Cao:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbpol.2024.122628.

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